Performance Analysis of IPv6/IPv4 Network Traffic Translation Mechanisms

Introduction and Literature Review

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Chapter 1

Introduction

1.1 Motivation

IPv4 and IPv6 coexist because network infrastructure, applications, and access services cannot be migrated simultaneously. An IPv6-capable network may still need to communicate with IPv4-only services, while an application that supports only IPv4 may operate across an IPv6-only access network. Dual Stack allows a host to use both protocol families, but it does not provide communication between IPv6-only and IPv4-only endpoints by itself. Such communication requires an appropriate transition mechanism.

NAT64 and MAP-T provide different forms of IPv4/IPv6 interoperability. Stateful NAT64 translates traffic from IPv6 clients to IPv4 servers while maintaining translation state; MAP-T uses algorithmic address and port mapping [2], [4]. When translation is performed on the sending host, packets undergo additional processing before they are forwarded. Depending on the implementation and workload, these operations may affect latency, throughput, and packet loss.

The measured performance also depends on the workload. Packet size, offered load, the number of active flows, and flow creation rate can influence packet-processing behaviour and the point at which performance begins to deteriorate. These workload parameters should therefore be varied alongside throughput and packet loss. Otherwise, a configuration tested under a lighter load may appear faster simply because it handles less traffic.

DNS resolution is another part of application-visible communication time. A name-based request may require a lookup before a connection can be established. DNS64 is relevant to NAT64 because it can synthesize an AAAA record from an A record, allowing an IPv6 client to address an IPv4-only server through the configured NAT64 prefix [3]. Cache state affects this stage: a valid cached answer may be returned without an upstream lookup, whereas a cache miss may require additional DNS exchanges.

A host-side DNS cache-clearing event does not necessarily affect packets in an already established flow. Its expected effect is primarily on subsequent name-based requests that cannot obtain a valid answer from another cache. It may therefore increase the time before a new connection can start, rather than the packet-processing delay of an existing data

flow. DNS lookup time, connection-establishment time, data-path latency, and complete transaction time must be distinguished when evaluating this effect.

For legacy applications with strict timing requirements, average performance is not the only relevant property. Large variations in request or response time may be problematic even when the average remains acceptable. This motivates an investigation of how transition mechanisms, DNS behaviour, and cache state affect both communication performance and its predictability.

This thesis combines a controlled comparison of sender-host configurations with an evaluation of an adaptive design. Dual Stack is the default mode. Following a detected host-side DNS cache-clearing event, the proposed design may direct eligible new requests to an IPv6-to-IPv4 NAT64 path and use an independent local DNS64 resolver. MAP-T is considered separately as an additional stateless path, subject to the addressing and packet-path requirements of the selected implementation. The study will test whether this arrangement reduces DNS-related variation and what additional delay or operational complexity it introduces; no improvement is assumed in advance.

1.2 Problem Statement

The central problem is how to maintain predictable application-visible communication time while providing IPv4/IPv6 interoperability on the sending host. Existing transition mechanisms provide different forms of connectivity, but their performance depends on implementation, topology, and workload. Measurements obtained from a dedicated gateway cannot automatically describe a translator placed directly in the data path of an application running on an endpoint.

DNS cache state creates a separate source of variation. If a valid record is unavailable in the relevant cache, a new name-based request may require DNS resolution before connection establishment can begin. This delay should not be confused with the forwarding or translation delay of packets in an established connection. Its contribution depends on resolver placement, cache state, record validity, and the way the application obtains its destination address.

The proposed adaptive design uses Dual Stack as its default mode and considers NAT64 after detection of a host-side DNS cache-clearing event. However, switching the packet path alone is insufficient to benefit from a retained DNS answer. The control mechanism must also ensure that the relevant application query reaches the independent DNS64 resolver and that the application can use the synthesized IPv6 address. The resolver's cache must remain independent of the host cache being cleared, and its answers must still be valid under DNS TTL rules. If these conditions are not met, the proposed switch may add complexity without reducing lookup time.

MAP-T raises a different set of constraints. Its stateless mapping rules are not DNS records and cannot resolve domain names. A DNS lookup may provide an address that is

then processed according to applicable MAP rules, but the mapping is valid only within the corresponding MAP configuration and deployment model. In addition, a conventional MAP-T deployment may rely on a Border Relay for external IPv4 reachability [4]. A sender-host MAP-T result can therefore be reported only after the actual packet path and endpoint addressing have been verified.

The research problem has two parts. First, the network-performance characteristics of native IPv4 and IPv6 operation, NAT64, and eligible MAP-T configurations must be compared under consistent conditions. Second, the adaptive design must be implemented and evaluated to determine whether event detection, DNS64 cache use, and path selection improve application-visible timing predictability enough to justify their additional delay and deployment constraints.

1.3 Aim and Research Questions

The aim of this research is to design, implement, and evaluate an adaptive IPv4/IPv6 communication mechanism based on existing transition technologies. Dual Stack will serve as the default mode. After a host-side DNS cache-clearing event is detected, the mechanism will consider directing eligible new IPv6-originated requests to IPv4-only destinations through NAT64, while using an independent local DNS64 resolver to provide valid synthesized answers. MAP-T will be evaluated as an additional stateless path where its sender-host applicability can be demonstrated.

The study will compare separate IPv4 and IPv6 Dual Stack baselines with NAT64 and MAP-T configurations that satisfy the deployment requirements. It will then evaluate the adaptive prototype. The main measurements will include DNS lookup time, connection-establishment time, complete transaction time and its variability, data-path latency, throughput, and packet loss.

The research is guided by the following questions:

RQ1: What network-performance characteristics are observed for IPv4 and IPv6 Dual Stack operation and for eligible sender-host NAT64 and MAP-T configurations?

RQ2: How do DNS resolution, cache state, and host-side DNS cache-clearing events affect lookup time, connection establishment, and complete transaction time?

RQ3: How can Dual Stack, NAT64, and an optional MAP-T path be coordinated in an adaptive architecture while preserving DNS record validity and MAP configuration validity?

RQ4: To what extent does the adaptive mechanism change transaction-time variability and data-path performance compared with the standalone configurations under different workloads?

RQ5: What additional delay and timing variability are introduced by event detection, path switching, DNS64 cache use, and optional MAP-T integration, and what deployment constraints affect the mechanism's suitability for legacy applications with strict timing requirements?

1.4 Proposed Approach and Scope

The work will begin with a baseline comparison. IPv4 and IPv6 traffic on a dual-stack host will be measured separately. Stateful NAT64 with DNS64 will be tested for the IPv6-client-to-IPv4-server scenario supported by the selected implementation. MAP-T will be included only if a concrete implementation can process the required traffic on the sending host under a valid endpoint and relay configuration. Dual Stack is a native protocol-coexistence configuration, not a translation mechanism, and will be treated as the reference case.

In the adaptive design, Dual Stack is the normal operating mode. A monitoring component detects a host-side DNS cache-clearing event and notifies the control module. For eligible new requests, the control module may then select the NAT64 path. This transition must include the DNS side of the operation: the relevant application must query the independent DNS64 resolver, receive a valid synthesized AAAA record when appropriate, and use that address for the NAT64 connection. A route change without the corresponding resolver and address-selection behaviour will not be treated as a complete implementation of the proposed mechanism.

The cache-clearing event must be observable. The implementation will use an operating-system, resolver, or cache-management event if the selected environment exposes one. If no reliable notification is available, the experiment will use a controlled cache-clearing trigger and clearly identify it as such; it will not assume that all operating systems provide a universal cache-clear notification. The resolver cache must be separate from the monitored host cache. Cached records will be used only while valid, with DNS TTL and DNS64 synthesis behaviour respected [3].

MAP-T will be treated as a separate stateless path rather than as an automatic stage in the NAT64 switch. Its mapping component will maintain the applicable MAP rules and perform address and port calculations only for traffic that belongs to the configured MAP domain. An optional preparation experiment may resolve selected domain names and calculate the relevant mapping information in advance, but DNS records and MAP rules remain distinct data with separate validity conditions. This preparation function is an experimental extension, not a DNS64 or MAP-T protocol feature and not an assumed source of improvement.

The MAP-T packet path requires particular care. RFC 7599 defines MAP Customer Edge and Border Relay roles and describes the addressing and translation behaviour of a MAP-T domain [4]. A configuration that relies on a remote Border Relay to perform

the translation being measured will not be reported as sender-host translation. The endpoint addressing, routes, and translation location will be documented. If no candidate MAP-T implementation meets the sender-host criterion, the result will be reported as an applicability limitation rather than replaced with an intermediate translator or another transition technology.

The experimental scope is limited to functions deployed on the sending host. A local network namespace or virtual interface may be used to place a translator in the path of locally generated application traffic, provided that the translation function remains on the sending machine and the additional virtual-network overhead is documented. Packet captures, routing information, implementation counters, and configuration records will be used to verify that the intended component actually processes the measured traffic [11], [12]. Existing connections will not be assumed to migrate safely between communication paths; the adaptive policy will apply to new requests unless an implementation-specific migration mechanism is separately demonstrated.

The return condition from NAT64 to Dual Stack will be defined as an explicit policy during implementation. The policy, its trigger, and the delay associated with switching will be documented and measured. This is necessary because an unspecified return rule would make the duration of the adaptive state—and therefore the comparison—dependent on an unrecorded implementation choice.

1.5 Prototype and Evaluation Plan

The prototype is expected to contain a host DNS-cache monitoring component, an adaptive control module, an independent local DNS64 resolver, and a MAP-T mapping component for eligible experiments. The optional mapping-preparation function will be evaluated separately from the core adaptive path. Logs will associate each measurement with the cache state, detected event, control decision, DNS activity, and selected communication path.

The test environment will contain a sending host, a receiving host, a controlled forwarding path, and the DNS services required by each configuration. The sending host will run the workload generator and the translation functions under test. The receiver will expose the address family required by each scenario. Before performance testing, the topology will be checked to ensure that no unrecorded intermediate translator or relay performs the measured translation.

The evaluation will proceed in stages. First, sender-host applicability will be verified for NAT64 and MAP-T, and baseline measurements will be collected for each eligible configuration. Second, the adaptive prototype will be tested with populated, cold, and deliberately cleared host-side DNS caches. Third, the effects of the independent DNS64 cache, MAP-T mapping, and optional background preparation will be isolated where the implementation permits. Switching overhead and the return policy will also be measured.

Each admitted configuration will be tested with DNS and without DNS. DNS-enabled tests will use name-based requests and include cold-cache and warm-cache conditions. DNS-free tests will use equivalent explicit-address requests to isolate the data path. Where the application protocol uses the hostname for virtual hosting or TLS certificate selection, those application semantics will be preserved in paired tests. DNS lookup, connection establishment, data transfer, and complete transaction time will be recorded separately.

The workload will vary packet size, offered load, the number of active flows, and flow creation rate. UDP traffic will be used for controlled offered-load measurements; TCP transfers may be added as complementary application-oriented tests. Network measurements will include data-path latency, throughput or application goodput, and packet loss. Latency distributions, including median and tail values, will be reported alongside averages.

The comparison will use the same test environment, software versions, endpoint settings, and measurement tools wherever the architecture allows. The experiment record will include the operating system and kernel, translator and resolver versions, interface configuration, MTU, offload settings, traffic parameters, cache-reset procedure, and run duration. RFC 8219 and RFC 9693 will provide methodological references; any adaptations will be documented, and the experiments will not be described as fully compliant unless the relevant requirements are met [1], [10].

The final analysis will determine whether the adaptive design reduces DNS-related variation in complete transaction time, whether it changes steady-state data-path performance, and what additional delay, timing variability, and operational complexity it introduces. Results will be interpreted for the tested implementations, topologies, and workloads rather than generalized to every implementation of the protocols.

1.6 Thesis Organization

The theoretical background will review IPv4/IPv6 transition technologies, DNS resolution, caching mechanisms, and existing performance studies. The methodology will define the experimental model, deployment constraints, workload parameters, measurement criteria, and evaluation procedure. The implementation discussion will describe the adaptive communication mechanism, including DNS cache monitoring, path selection, the local DNS64 resolver, and MAP-T integration. The evaluation will present the experimental results, performance comparison, and limitations, followed by conclusions and directions for future work.

Chapter 2

Literature Review

2.1 Review Scope and Comparison Criteria

The reviewed sources serve different purposes. Standards documents specify protocol behaviour and deployment roles; survey and classification studies describe the wider transition-technology landscape; and benchmarking publications provide methods for measuring performance and scalability. Implementation documentation is used to identify practical constraints that may affect whether a translator can process traffic generated on the sending host. These source types provide complementary evidence and should not be treated as interchangeable.

The comparison in this thesis focuses on four aspects: the function and location of translation, the presence or absence of per-flow state, DNS requirements, and the network-performance effects observable on the sending host. Latency, throughput, and packet loss will be considered alongside DNS lookup, connection-establishment, and complete transaction times. This separation is important because name resolution and packet translation affect different stages of an application request.

2.2 Dual Stack and IPv4/IPv6 Coexistence

Dual Stack allows a host or network device to operate with both IPv4 and IPv6. It provides a useful reference configuration because it does not require packet translation between the two protocol families on the tested path. The IPv4 and IPv6 cases should nevertheless be measured separately: address selection, routes, headers, and effective payload sizes may differ. Combining both protocol families into a single result would obscure these differences.

Lencse and Kadobayashi provide a broad survey and classification of IPv6 transition technologies, including their architectural characteristics and security considerations [7]. Hamdou and colleagues apply a multiple-criteria decision-making approach to the classification of transition mechanisms [6]. These works help position Dual Stack, stateful

translation, and stateless mapping within the wider transition landscape. Their classifications do not replace direct measurements of latency, throughput, or packet loss for a particular sender-host implementation.

Accordingly, this thesis treats native IPv4 and native IPv6 communication on the dual-stack host as two distinct reference cases. They provide a baseline for the host networking stack and the measurement environment before translation functions are introduced.

2.3 Stateful NAT64 and DNS64

Stateful NAT64 enables IPv6 clients to initiate communication with IPv4 servers by translating packet headers and maintaining state for supported flows [2]. Its performance may therefore depend on the number of active mappings, the rate of connection creation and termination, and the behaviour of the implementation's state tables. These factors motivate tests that vary both concurrent flow count and flow turnover.

DNS64 complements NAT64 by synthesizing AAAA records from A records when a name does not provide a suitable AAAA record [3]. The synthesized address represents the IPv4 destination through the prefix configured for the NAT64 environment. DNS64 and NAT64 perform different functions: DNS64 provides name-resolution and address-synthesis behaviour, whereas NAT64 translates packets. Measurements that use both should therefore retain separate observations of lookup time and data-path performance.

RFC 6147 also makes cache validity relevant to the proposed design. A synthesized answer depends on the validity of the source DNS data and must not be treated as permanently reusable [3]. An independent DNS64 resolver can retain an answer after the host-side cache has been cleared only if its own cache has not been cleared and the answer remains valid. Applications must also send their queries to that resolver and use the synthesized address. Thus, an independent cache is a possible way to reduce repeated upstream resolution, but its presence alone does not guarantee a faster application request.

2.4 MAP-T and Stateless Address Mapping

MAP-T provides stateless IPv4/IPv6 translation using algorithmic address and port mapping rules [4]. Unlike stateful NAT64, it does not maintain per-flow translation state in the same manner; the mapping is derived from configured rules and packet address and port information. This architectural difference motivates testing the mechanisms under workloads that vary both active-flow count and flow creation rate.

RFC 7599 defines MAP Customer Edge and MAP Border Relay roles and specifies the mapping behaviour within a MAP-T domain [4]. The Border Relay provides connectivity between the MAP-T domain and external IPv4 networks. The specification also describes communication with suitably addressed native IPv6 servers in a MAP-T domain. Consequently, the endpoint addressing and role of any Border Relay must be stated for every

MAP-T experiment. The use of MAP-T as a sender-host translation result cannot be inferred merely from the presence of a MAP-T implementation on the sending machine.

MAP rules must also be distinguished from DNS records. DNS can provide an address for a domain name; MAP rules define how addresses and ports are represented within a configured MAP domain. A proposed preparation component may obtain DNS information for selected names and then calculate applicable mapping information, but it cannot use MAP rules to resolve a domain name or create rules that are absent from the configuration. DNS records and MAP rules have separate validity and update conditions. Background preparation is therefore an optional experiment rather than a standard feature of MAP-T or DNS64.

2.5 Performance Benchmarking of Transition Mechanisms

RFC 8219 provides a benchmarking methodology for IPv6 transition technologies [1]. It emphasizes defined test setups, traffic characteristics, and repeatable measurements. It is a methodological foundation for this thesis, but using selected metrics from the RFC does not by itself make an experiment compliant. Any differences in topology, traffic generation, timing, or repetition will be recorded.

Lencse, Shima, and Cho present a benchmarking methodology for stateful NAT64 gateways, addressing the evaluation of gateway performance and scalability [8]. RFC 9693 provides a related methodology for benchmarking stateful NATxy gateways [10]. These sources are relevant to the measurement of stateful translation, including the need to consider connection establishment and state-table behaviour. Their gateway-oriented results cannot be transferred directly to an endpoint deployment, where the translator is placed in the same packet path as the application and workload generator.

2.6 Empirical Studies of MAP-T and Translation Scalability

Lencse and Bazsó examine benchmarking methodology for IPv4-as-a-Service technologies and compare the scalability of Jool implementations of 464XLAT and MAP-T [5]. The MAP-T-related methodology and workload considerations are relevant to the present study. Their 464XLAT experiments are background literature only: 464XLAT is not included in the implementation plan or as a reserve technology for this thesis.

Al-hamadani and Lencse introduce Maptperf, a tester for benchmarking MAP-T Border Relay routers in accordance with RFC 8219 [9]. This work provides a reference for evaluating MAP-T relay performance, but it does not establish that a given MAP-T implementation can perform the required translation locally on a sending host. Its results

must therefore be interpreted in light of the different measurement location and topology.

Hamdou and colleagues provide a multi-criteria classification of transition mechanisms [6]. Their work offers a structured overview of factors such as performance, deployment, and routing considerations. An aggregate classification, however, cannot replace measurements of a specific implementation under controlled traffic loads. The present study complements this literature with a sender-host comparison and explicit verification of translation placement.

2.7 DNS Caching and Transaction Performance

The standards and benchmarking sources establish the roles of DNS64 and NAT64, but the performance evaluation in this thesis must also distinguish the stages of a name-based request. A cache hit, a cache miss, and a deliberately cleared host cache can produce different lookup behaviour. Their effects should be measured at the resolver and application level rather than attributed automatically to packet translation.

For this reason, the experiment will pair DNS-enabled name-based requests with equivalent explicit-address requests. The former will capture the combined effects of lookup, connection establishment, and data transfer; the latter will help isolate the data path. Cold-cache and warm-cache conditions will be defined explicitly. The complete transaction time will be reported alongside its constituent stages so that any reduction in lookup delay can be distinguished from a change in steady-state packet processing.

The proposed independent DNS64 cache extends this evaluation by testing whether a valid resolver-side answer remains available after the monitored host cache is cleared. The result depends on resolver configuration, cache independence, TTL validity, and application address selection. It will therefore be treated as an experimental hypothesis, not as a benefit guaranteed by DNS64 itself.

2.8 Host-Side Translation Constraints

The location of translation is essential to the scope of this thesis. A translator installed on a sending host does not necessarily process packets generated by applications in the host's ordinary network namespace. The actual packet path depends on the implementation's hooks, interfaces, routes, and namespace configuration.

The Jool documentation describes a node-based translation arrangement in which a translator is placed between an application and the network interface using network namespaces and virtual interfaces [11]. Its FAQ also describes a case in which a Netfilter Jool instance attached to an incoming-packet hook does not automatically intercept packets generated in its own network namespace [12]. These implementation-specific details illustrate why installation alone is insufficient evidence of sender-host translation.

The experiment will therefore verify the packet path using captures, routing information,

implementation counters, and configuration records. The same principle applies to MAP-T: the CE/BR arrangement, endpoint addressing, and location of translation must be documented. Processing performed by a remote Border Relay will not be counted as sender-host processing. If a candidate implementation cannot satisfy the placement requirement, that limitation will be reported explicitly.

2.9 Research Position and Identified Gap

The reviewed standards define the behaviour and deployment roles of stateful NAT64, DNS64, and MAP-T [2]–[4]. Survey and classification studies describe the wider IPv4/IPv6 transition-technology landscape [6], [7]. Existing benchmarking work provides methods for measuring the performance of transition technologies, MAP-T Border Relays, and stateful NAT64 or NATxy gateways [1], [5], [8]–[10]. Implementation documentation further shows that sender-side packet interception depends on the concrete software and network configuration [11], [12].

These studies characterize important performance properties of individual transition mechanisms and implementations, including latency, throughput, and scalability. However, within the reviewed material, the problem of increased application response time caused by the additional packet-processing and forwarding overhead of transition mechanisms is not directly addressed through an adaptive communication scheme. In particular, the reviewed works do not evaluate how the selection of a communication path can be coordinated with DNS cache state to reduce the time an application waits for a response while preserving IPv4/IPv6 connectivity. This problem is important for applications with strict timing requirements, where both response time and its variability can affect system operation.

To address this problem, this thesis proposes an adaptive scheme intended to optimize application-visible response time by balancing DNS-resolution delay against the processing overhead of the selected communication path. Dual Stack serves as the default mode, allowing native communication to be used where it is available. When a host-side DNS cache-clearing event is detected, the mechanism may direct eligible new IPv6-originated requests to IPv4-only destinations through NAT64 and use an independent local DNS64 resolver. The independent resolver is intended to retain valid DNS information across the host-side cache-clearing event, provided that the application is directed to use it and the cached records remain valid under their TTL values. MAP-T is considered as an optional stateless translation path and mapping-preparation component only where the addressing configuration, endpoint requirements, and actual packet path support its use. Its mapping rules remain separate from DNS records and do not replace DNS resolution.

The proposed scheme is therefore an optimization approach to reducing the overall time between an application request and the corresponding response, rather than an assumption that one transition mechanism is always faster than another. Its effectiveness will be established experimentally by measuring DNS lookup time, connection-establishment time,

complete transaction time and its variability, data-path latency, throughput, and packet loss. The evaluation will also quantify the overhead introduced by event detection, path switching, DNS64 cache use, and optional MAP-T integration. Conclusions will be limited to the implementations, topologies, and workloads actually tested.

References

[1] M. Georgescu, L. Pislaru, and G. Lencse, *Benchmarking Methodology for IPv6 Transition Technologies*, RFC 8219, Aug. 2017. https://doi.org/10.17487/RFC8219.

[2] M. Bagnulo, P. Matthews, and I. van Beijnum, *Stateful NAT64: Network Address and Protocol Translation from IPv6 Clients to IPv4 Servers*, RFC 6146, Apr. 2011. https://doi.org/10.17487/RFC6146.

[3] M. Bagnulo, A. Sullivan, P. Matthews, and I. van Beijnum, *DNS64: DNS Extensions for Network Address Translation from IPv6 Clients to IPv4 Servers*, RFC 6147, Apr. 2011. https://doi.org/10.17487/RFC6147.

[4] X. Li, C. Bao, W. Dec, O. Troan, S. Matsushima, and T. Murakami, *Mapping of Address and Port using Translation (MAP-T)*, RFC 7599, Jul. 2015. https://doi.org/10.17487/RFC7599.

[5] G. Lencse and Á. Bazsó, "Benchmarking methodology for IPv4aaS technologies: Comparison of the scalability of the Jool implementation of 464XLAT and MAP-T," *Computer Communications*, vol. 219, pp. 243–258, 2024. https://doi.org/10.1016/j.comcom.2024.03.007.

[6] M. Hamdou, M. El Haloui, A. El Ksimi, and B. Ettaki, "Classification of IPv6 Transition Mechanisms using Multiple-Criteria Decision-Making," *Engineering, Technology & Applied Science Research*, vol. 15, no. 3, pp. 22960–22968, 2025. https://doi.org/10.48084/etasr.10222.

[7] G. Lencse and Y. Kadobayashi, "Comprehensive Survey of IPv6 Transition Technologies: A Subjective Classification for Security Analysis," *IEICE Transactions on Communications*, vol. E102-B, no. 10, pp. 2021–2035, 2019. https://doi.org/10.1587/transcom.2018EBR0002.

[8] G. Lencse, K. Shima, and K. Cho, "Benchmarking methodology for stateful NAT64 gateways," *Computer Communications*, vol. 210, pp. 256–272, 2023. https://doi.org/10.1016/j.comcom.2023.08.009.

[9] A. Al-hamadani and G. Lencse, "Maptperf: An RFC 8219 compliant tester for benchmarking MAP-T border relay routers," *Computer Networks*, vol. 257, article 111012, 2025. https://doi.org/10.1016/j.comnet.2024.111012.

[10] G. Lencse and K. Shima, *Benchmarking Methodology for Stateful NATxy Gateways*, RFC 9693, Jan. 2025. https://doi.org/10.17487/RFC9693.

[11] Jool Project, "Node-Based Translation," implementation documentation. Accessed Oct. 12, 2026. https://www.jool.mx/en/node-based-translation.html.

[12] Jool Project, "FAQ: Why is my ping not working?" implementation documentation. Accessed Oct. 12, 2026. https://www.jool.mx/en/faq.html.